

WENYUAN LI

wenyuan.li@usc.edu

KAP 424A, 3620 S. Vermont Ave, Los Angeles, CA, U.S.

EDUCATION

Department of Mathematics, Northwestern University 09/2018-06/2023

Ph.D. in Mathematics

Advisor: Prof. Emmy Murphy & Prof. Eric Zaslow.

Thesis: Some functoriality results for microlocal sheaves over Legendrians and Lagrangians.

Department of Mathematics, Princeton University 09/2021-06/2022

Visiting graduate student

Department of Mathematics, Peking University 09/2015-07/2018

B.A. in Mathematics

Advisor: Prof. Huijun Fan,

Undergraduate thesis: Geometry of Lagrangian submanifolds in symplectic manifolds.

Department of Psychology, Peking University 09/2014-09/2015

EMPLOYMENT

Department of Mathematics, University of Southern California 08/2023-05/2026

Assistant Professor (RTPC) in Mathematics

RESEARCH INTEREST

Symplectic topology, contact topology, microlocal sheaf theory, and their connections to cluster algebras, mirror symmetry, noncommutative geometry, algebraic K-theory and persistence modules.

AWARDS AND GRANTS

AMS-Simons Travel Grant

07/2024-06/2026

PAPERS AND PREPRINTS

1. Lagrangian cobordism functor in microlocal sheaf theory I,

J. Topol. 16.3 (2023): 1113-1166, arXiv:2108.10914.

2. Existence of generating families on Lagrangian cobordisms,

Math. Ann. 390 (2024), no.4, 5471-5494, arXiv:2308.05727.

3. Lagrangian cobordism and shadow distance in Tamarkin category,

joint with Tomohiro Asano and Yuichi Ike, *Sel. Math. New Ser.* 31, 45 (2025), arXiv:2312.14429.

4. Duality and kernels in microlocal geometry,

joint with Christopher Kuo, *Int. Math. Res. Not.* (2025), no.6, 1-34, arXiv:2405.15211.

5. Lagrangian cobordism functor in microlocal sheaf theory II,

J. Sympl. Geom. 23 no.3 (2025): 599-672, arXiv:2308.05089.

6. Positive microlocal holonomies and globally regular,

joint with Roger Casals, *Comm. Amer. Math. Soc.*, 6 (2026), 492-535, arXiv:2409.07435.

7. Estimating Reeb chords using microlocal sheaf theory,

39pp, to appear in *Algebr. Geom. Topol.*, arXiv:2106.04079.

- 8. Conjugate fillings and Legendrian weaves,**
joint with Roger Casals, 100pp, submitted, arXiv:2210.02039.
- 9. Spherical adjunction and Serre functor from microlocalization,**
joint with Christopher Kuo, 62pp, submitted, arXiv:2210.06643.
- 10. Relative Calabi-Yau structure on microlocalization,**
joint with Christopher Kuo, 50pp, submitted, arXiv:2408.04085.
- 11. Decompositions of augmentation varieties via weaves and rulings,**
joint with Johan Asplund, Orsola Capovilla-Searle, James Hughes, Caitlin Levenson and Angela Wu, 88pp, submitted, arXiv:2508.20226.
- 12. C^0 -rigidity of Legendrians and coisotropics via sheaf quantization,**
joint with Tomohiro Asano, Yuichi Ike and Christopher Kuo, 45pp, submitted,
arXiv:2510.01746.
- 13. Microsheaf composition of Lagrangian correspondences,**
joint with David Nadler and Vivek Shende, 105pp, arXiv:2511.19814.

RESEARCH TALKS

Bonn Symplectic Geometry Seminar Lagrangian correspondence for microlocal sheaves	<i>Jun 22, 2026</i>
Symplectic Zoominar (CRM-Montréal, Princeton/IAS, Tel Aviv, and Paris) C^0 -rigidity in contact topology via microlocal sheaf theory	<i>May 29, 2026</i>
Northwestern Geometry/Physics Seminar Relative Calabi–Yau structure on microlocal sheaves	<i>May 12, 2026</i>
MIT Symplectic Geometry Seminar Lagrangian correspondence for microlocal sheaves	<i>Feb 19, 2026</i>
Stanford Symplectic Geometry Seminar C^0 -rigidity of Legendrian and coisotropic submanifolds	<i>Jan 12, 2026</i>
PU/IAS Symplectic Geometry Seminar Lagrangian correspondence for microlocal sheaves	<i>Oct 21, 2025</i>
Conformal Dynamics and Geometry in Bordeaux (short talk) C^0 -rigidity of Legendrian submanifolds	<i>Aug 26, 2025</i>
Loo-Keng Hua Youth Forum (topology session), AMSS C^0 -rigidity of Legendrian submanifolds	<i>Aug 11, 2025</i>
Kyoto University Symplectic Geometry Seminar Lagrangian correspondence in topological Fukaya categories	<i>Jul 11, 2025</i>
CBMS Summer School, Legendrian Links and the Microlocal Theory of Sheaves Structural results for microlocal sheaves for Legendrians	<i>Jun 10, 2025</i>
AMS Western Sectional Meeting, Recent Developments in Contact and Symplectic Geometry (short talk) Global regular functions on the moduli of Lagrangian fillings	<i>May 04, 2025</i>
Geometric Representation Theory Seminar, UC Berkeley Regular functions on the moduli of Lagrangian fillings	<i>Apr 17, 2025</i>

BIRS Workshop on Symplectic Geometry, Representation Theory and Cluster Algebras (short talk)	
Symplectic geometric construction of cluster variables on braid varieties	<i>Apr 01, 2025</i>
University of Toronto Symplectic Geometry Seminar	
Relative Calabi-Yau structure on microlocal sheaves	<i>Mar 17, 2025</i>
AMS Southeastern Sectional Meeting, Invariants in Geometric Topology: Low Dimensions and Beyond (short talk)	
Global regular functions on the moduli of Lagrangian fillings	<i>Mar 08, 2025</i>
2024 Annual International Congress of Chinese Mathematicians	
Structural results for microlocal sheaves in symplectic geometry	<i>Jan 03, 2025</i>
The 1st Cross-strait Symplectic Geometry Conference, USTC	
Lagrangian cobordisms and microlocal sheaf theory	<i>Dec 28, 2024</i>
Center for Quantum Mathematics Seminar, University of Southern Denmark	
Relative Calabi-Yau structures for microlocal sheaves	<i>Dec 17, 2024</i>
AMS Western Sectional Meeting, Non-commutative algebras in representation theory and topology (short talk)	
Relative Calabi-Yau structures for microlocal sheaves	<i>Oct 26, 2024</i>
Shanghai Jiao Tong University Topology Seminar	
Structural results of microlocal sheaf categories for Legendrians	<i>Jul 16, 2024</i>
China Academy of Science AMSS Topology Seminar	
Relative Calabi-Yau structures for microlocal sheaves/Fukaya categories	<i>Jul 10, 2024</i>
UC Riverside Geometry & Topology Seminar	
Relative Calabi-Yau structures for microlocal sheaves/Fukaya categories	<i>May 10, 2024</i>
UC Davis Algebraic Geometry Seminar	
Relative Calabi-Yau structures for microlocal sheaves	<i>Mar 13, 2024</i>
Stanford Symplectic Geometry Seminar	
Generating families on Lagrangian cobordisms	<i>Mar 11, 2024</i>
Duke Geometry & Topology Seminar	
Structural results of microlocal sheaf categories for Legendrians	<i>Feb 19, 2024</i>
The 15th International Symposium of Natural Sciences, INU (online)	
Generating families on Lagrangian cobordisms	<i>Oct 30, 2023</i>
Claremont Topology Seminar	
Generating families on Lagrangian cobordisms	<i>Oct 24, 2023</i>
LSU Geometry & Topology Seminar	
Conjugate fillings and Legendrian weaves	<i>Oct 11, 2023</i>
Caltech/USC Algebra & Geometry Seminar	
Lagrangian cobordism functor in microlocal sheaf theory	<i>Oct 05, 2023</i>
USC Geometry & Topology Seminar	
Lagrangian cobordism functor in microlocal sheaf theory	<i>Aug 21, 2023</i>
Peking University Algebra & Geometry Mini Forum	
Duality and spherical adjunction from microlocalization	<i>Aug 01, 2023</i>

Georgia Topology Conference 2023

Lightning talk - Conjugate fillings and Legendrian weaves *May 25, 2023*

Tianjin University Geometry & Topology Seminar (online)

Conjugate fillings and Legendrian weaves *Dec 16, 2022*

The 2nd Global Youth Forum of IGP, USTC (online)

Lagrangian cobordism functor in sheaf theory *Dec 14, 2022*

Boston College Geometry & Topology Seminar

Conjugate fillings and Legendrian weaves *Dec 01, 2022*

Northwestern Geometry & Physics Seminar

Duality and spherical adjunction for microlocal sheaves *Nov 03, 2022*

Columbia Symplectic Geometry, Gauge Theory & Categorification Seminar

Duality and spherical adjunction for microlocal sheaves *Oct 14, 2022*

UIUC Symplectic & Poisson Geometry Seminar

Conjugate fillings and Legendrian weaves *Oct 03, 2022*

Uppsala Symplectic Geometry Seminar (online)

Lagrangian cobordism functor in sheaf theory *Sep 29, 2022*

Michigan State University Geometry & Topology Seminar

Lagrangian cobordism functor in microlocal sheaf theory *Mar 01, 2022*

Symplectic Zoominar (CRM-Montréal, Princeton/IAS, Tel Aviv, and Paris)

Short talk - Estimating Reeb chords using microlocal sheaf theory *Dec 17, 2021*

EXPOSITORY TALKS

Symplectothon Spring 2026 organized by Kyler Siegel and Sheel Ganatra

Introduction to microlocal sheaf theory *Apr 17, 2026*

Sheaf quantization of exact Lagrangians *Apr 18, 2026*

Microlocal Sheaf Theory Learning Seminar organized by Chris Kuo and Wenyan Li

Overview of microlocal sheaf theory and its applications *Jan 31, 2025*

Microlocalization and microsheaves *Mar 28, 2025*

Legendrian Contact Homology Learning Seminar organized by Roman Krutowski

Lagrangian cobordism functor in microlocal sheaf theory *Nov 17, 2023*

Graduate Student Seminar at Northwestern University

Families of functions and Lagrangian submanifolds *May 12, 2023*

Symplectic ‘N’ Contact Topology Early Career Workshop

Exotic Weinstein structure with many closed Lagrangians *Aug 09, 2022*

Virtual Kylerec Student Seminar 2021

An introduction to loose Legendrians in high dimensions *Jun 15, 2021*

Graduate Student Seminar at Northwestern University

Meromorphic differential equations and Riemann-Hilbert correspondence *Feb 19, 2021*

Graduate Student Seminar at Northwestern University

Does symplectic topology of T^*M remember the smooth topology of M ? *Nov 13, 2020*

Graduate Student Seminar at Northwestern University

Asymptotically flat symplectic manifolds

Apr 19, 2019

Mirror Symmetry Seminar organized by Professor Huijun Fan & Professor Bo Dai

Keating's Lagrangian tori in four-dimensional Milnor fibers

Lagrangian surgery and computations of Maslov index

Oct, 2017

TEACHING EXPERIENCE

MATH 425a, Fundamental concepts of real analysis, instructor

Spring 2026

MATH 225, Differential equations and linear algebra, instructor

Fall 2025

MATH 535a, Differential geometry, instructor

Spring 2025

MATH 225, Differential equations and linear algebra, instructor

Fall 2024

MATH 126g, Calculus II, instructor

Spring 2024

MATH 126g, Calculus II, instructor

Fall 2023

GRE Math Sub preparation, Causeway Postbaccalaureate Certificate Program, instructor

Summer 2022

MATH 300-BR, Foundations of higher mathematics, TA

Summer 2022

MATH 290-3, MENU Linear algebra and multivariable calculus III, TA

Spring 2021

MATH 330-3, Abstract algebra III, TA

Spring 2021

MATH 290-2, MENU Linear algebra and multivariable calculus II, TA

Winter 2021

MATH 330-2, Abstract algebra II, TA

Winter 2021

MATH 290-1, MENU Linear algebra and multivariable calculus I, TA

Fall 2020

MATH 330-1, Abstract algebra I, TA

Fall 2020

MATH 230-1, Multi variable differential calculus, TA

Spring 2020

MATH 230-1, Multi variable differential calculus, TA

Winter 2020

MATH 220-1, Single variable differential calculus, TA

Fall 2019

SERVICE AND OUTREACH

Coorganizer of learning seminar on microlocal sheaf theory

Spring 2025

Coorganizer of AMS Special Session on Symplectic Geometry and Dynamics

Spring 2025

Coorganizer of AMS Special Session on Geometry and Topology of Contact and Symplectic Manifolds

Fall 2024

Coorganizer of Los Angeles Workshop on Representation and Geometry

Summer 2024

Coorganizer of Caltech/USC joint Algebra & Geometry Seminar

Fall 2024–Spring 2026

Coorganizer of USC Geometry & Topology Seminar

Fall 2023–Spring 2026

AWM mentoring program

Fall 2024–Spring 2025

Northwestern Association for Women in Mathematics mentoring program

Spring 2021

Instructor of UCLA Math Circle Advanced III

Fall 2024–Spring 2025

Directed reading program at Northwestern University

Spring 2023

Directed reading program at Northwestern University

Spring 2021

Instructor of Northwestern Putnam training sessions

Fall 2019

Instructor of Northwestern Putnam training sessions

Fall 2018

Qualifying exam committee member for Jonathan Michala (USC), Haosen Wu (USC), Ángel Manuel Rodríguez López (UC Davis), David O'Connor (USC), Boxi Hao (USC), Yijie Pan (USC), Baran Cetin (USC)

Referee (including quick opinions) for *Geom. Funct. Anal.*, *Geom. & Topol.*, *J. Topol.*, *Int. Math Res. Not.*, *Pacific J. Math.*, *Rend. Sem. Mat. Univ. Padova*

Reviewer for ZbMATH Open and MathReview.